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Electron Tunneling

FYSB22

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1 Theory

In the potential and quantum world, barriers do not work in the same way as they do in the Newtonian world. When a quantum particle encounters a barrier with higher potential energy than the kinetic energy of the particle some part of the particle can tunnel through the barrier. We can label this transmission probability as $T(E, 0)$ and the current of particle transmission as I.

One example of potential barriers can be two different materials with different energy gap between conduction and valence band, results in a potential difference in the material. Two such material with different conductivity is AlGaAs and GaAs [1]. Figure 1 show a setup of of one potential barrier and 2 show a setup of two potential barriers.

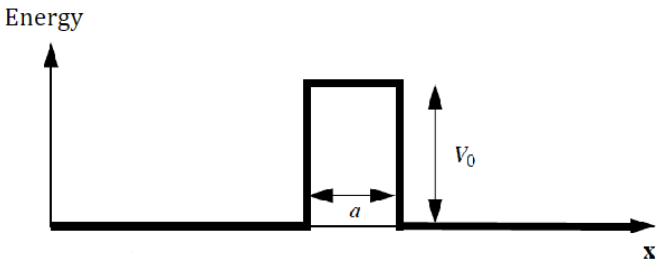


Figure 1: A drawing of a single barrier. [1]

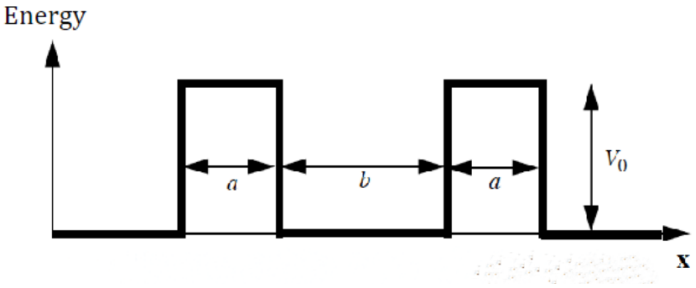


Figure 2: A drawing of a double barrier. [1]

In this lab four different samples, labeled A, B, C and D, were used. Their difference was that some had one layer of AlGaAs interposed between two plaques of GaAs whilst other had two layers of AlGaAs interposed between three plaques of GaAs. This can be seen in Figure 3 and Figure 4 where they are named single- and double barrier.

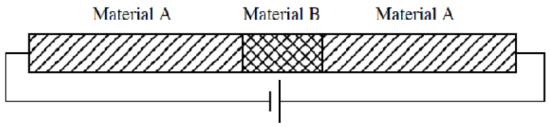


Figure 3: A schematic representation of the single barrier. [1]

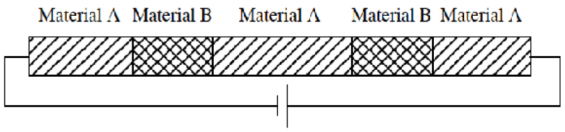


Figure 4: A schematic representation of the double barrier. [1]

When solving the time independent Schrödinger equation of one dimension we can find the transmission probability for a single barrier as

$$T(E,0) \approx e^{-2a\sqrt{2m^*(V_0-E)\hbar^2}}[1]. \tag{1}$$

In this equation a is the width of the barrier, e is Euler's constant, m^* is the effective mass in the material, V_0 is the potential of the barrier and E is the energy of the particle and $\hbar$ is Planck's constant. We can introduce

$$k = \sqrt{2m^*(V_0 - E)/\hbar^2} \tag{2}$$

in Equation (1) to make it look simpler [1]. The effective mass of an electron in GaAs is $m^* = 0.67m_e$ where m_e is the mass of an electron [1]. We can furthermore approximate $E \approx 0\,\text{eV}$ from the Equation (1) if the system is in a very low-temperature, that is the kinetic energy is approximated as zero [1]. Since we apply a bias voltage the barrier height will change as $V_0 \to V_0 - e\gamma U/2$, where e is the elementary charge, U is the bias voltage, and γ is a scaling factor of the potential bias. Since there is a potential drop in the circuit, the voltage we apply on barrier is not the voltage we measure, since we measure the voltage on the entire device. Thus, the scaling factor $\gamma = 0.3$ corresponds to a potential drop over the barrier [1]. Thus

$$k = \sqrt{2m^*(V_0 - e\gamma U/2)/\hbar^2} \tag{3}$$

This shift in the potential can be observed in Figure 5.

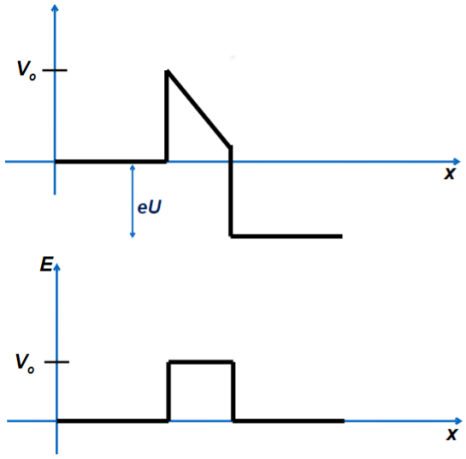


Figure 5: A picture showing the real potential of single barrier [1].

The same reasoning applies to the double barrier. Thus a bias will shift the potential barrier here as well, which can bee seen in Figure 6 .

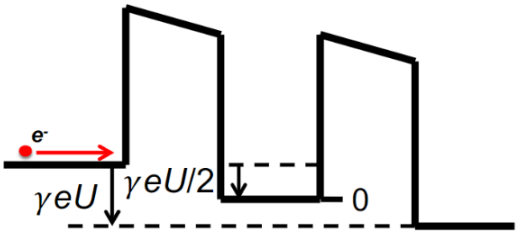


Figure 6: A picture showing the real potential of a double barrier. The x-axis has the dimension length, and the y-axis has the dimension energy [1].

Since the current is proportional to the transmission probability, we make the expression linear we can write Equation (1) linearly with the current I as

$$\ln(I) \approx a(-2k). \tag{4}$$

When there is a double barrier in the system the solution becomes more complicated. The transmission probability is

$$T(E,0) = \frac{T_B^2}{T_B^2 + 2(2 - \cos[2bk(E)])}, \tag{5}$$

where T_B is the transmission through each barrier independent of the energy and the wave

number

$$k(E) = \sqrt{2m^*E/\hbar^2}. \tag{6}$$

From Equation (5) one can tell that there will be a peak when the $\cos[2bk(E)]$ becomes zero. Thus we have resonance in the system meaning

$$U = U_{resonance}. \tag{7}$$

Here we would have a local current maxima in the system. In this first order local maxima we have the expression

$$E_1 = \frac{1}{2}\gamma eU = \frac{\hbar^2\pi^2}{2m^*b^2} \tag{8}$$

where E_1 is the energy in the ground state between the barriers, γ is the scaling factor of the potential bias, U is the potential bias and b is the distance between the barriers as in Figure 2 [1].

2 Method

To begin with, the circuit needs to be built. The top of the sample was connected to a power supply, and the bottom was connected to a ammeter in series and parallel connected to a voltmeter. The ammeter and the voltmeter was also connected to the power supply. This can be observed in Figure 7. Generic cables was used between the the supply, the voltmeter, and the ammeter. Coaxial cables was used between the sample and the supply and the ammeter.

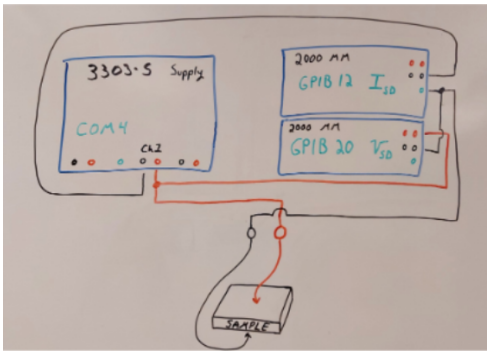


Figure 7: The circuit diagram. [1]

The sample consisted of a hetero-structure semiconductor. The material was GaAs and AlGaAs. The sample was submerged in liquid nitrogen and wired to the circuit. For the four different samples around 42 different data points were taken between 0 V to 1 V. To change the bias in the circuit a computer program was used, where one had to turn a knob to get a certain bias in the system. The data was downloaded to the computer for each sample.

3 Results

3.1 Sample A

When viewing Figure 8 we see a shape which looks like what the shape of a double barrier would look like. One can also observe a peak $U = 271.4\,\mathrm{mV}$.

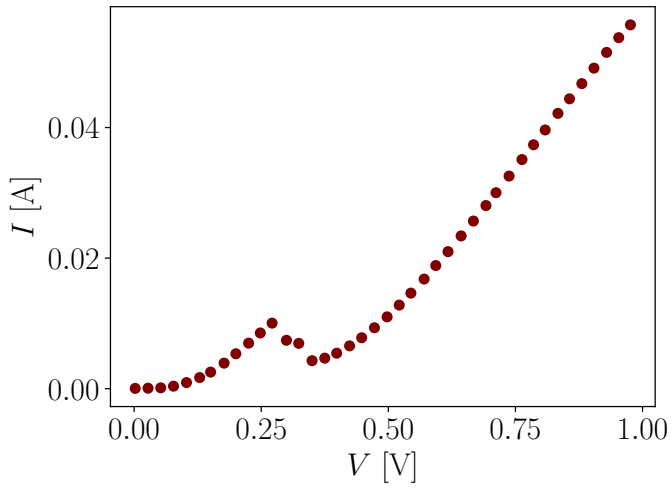


Figure 8: A scatter plot of the data collected from sample A.

3.2 Sample B

When viewing Figure 9 we see a shape witch looks like what on the shape of a double barrier. One can also observe a peak $U = 614.3\,\mathrm{m\,V}$.

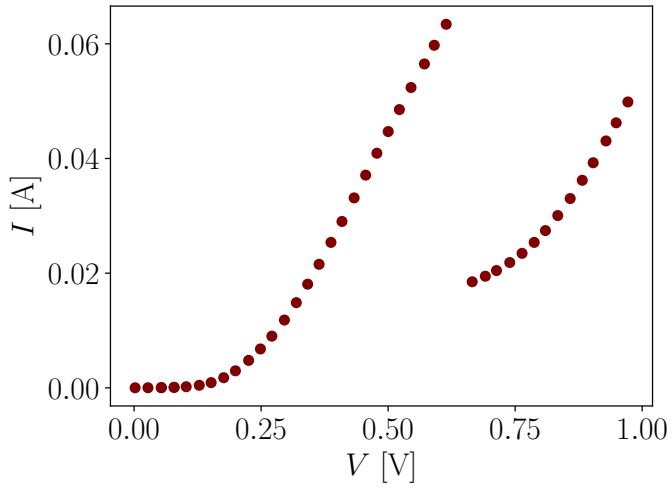


Figure 9: A scatter plot of the data collected from sample B.

3.3 Sample C

The plot obtained in Figure 13 is linear plot, with no peaks.

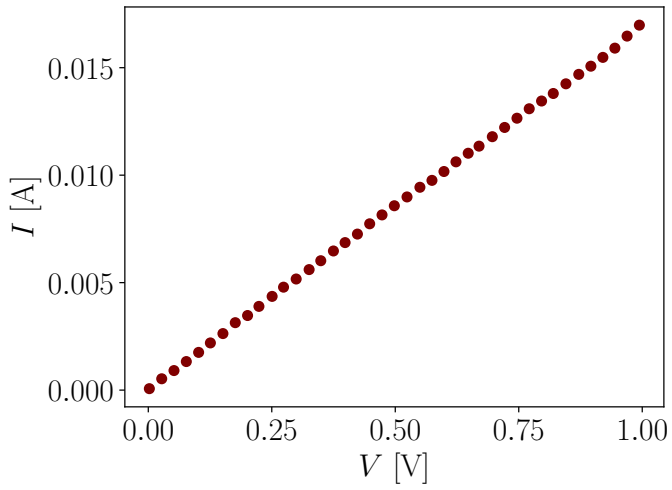


Figure 10: A scatter plot of the data collected from sample C.

3.4 Sample D

The plot in Figure 14 seems to be a exponential curve, with no visible peaks.

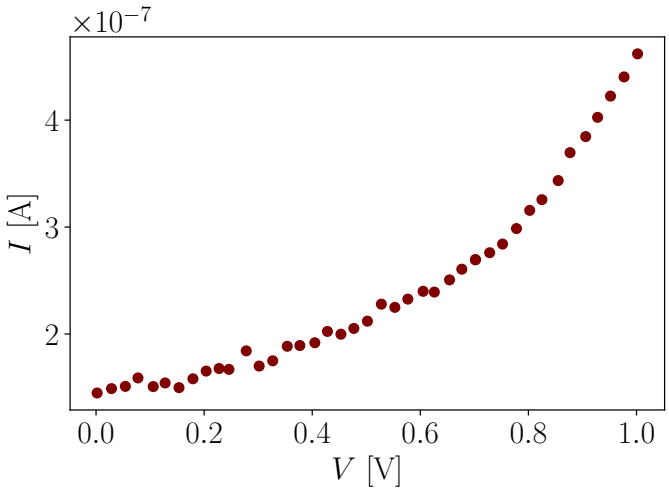


Figure 11: A scatter plot of the data collected from sample D.

4 Analysis

The data output files from the lab software always gave four significant figures. This implies that the answer should also be given with four significant figures.

4.1 Sample A

When viewing Figure 8 and comparing it with Equation (5) we can see that this sample is a double barrier. We can therefore apply Equation (8) to the local maxima of the scatter plot. In this local maxima, we know that the voltage is equal to that of the resonance voltage as in Equation (7). From Figure 12 we can tell that $U = 271.4\,\text{mV}$. From this we apply Equation (8) and get that the distance $b_A = 11.74\,\text{nm}$.

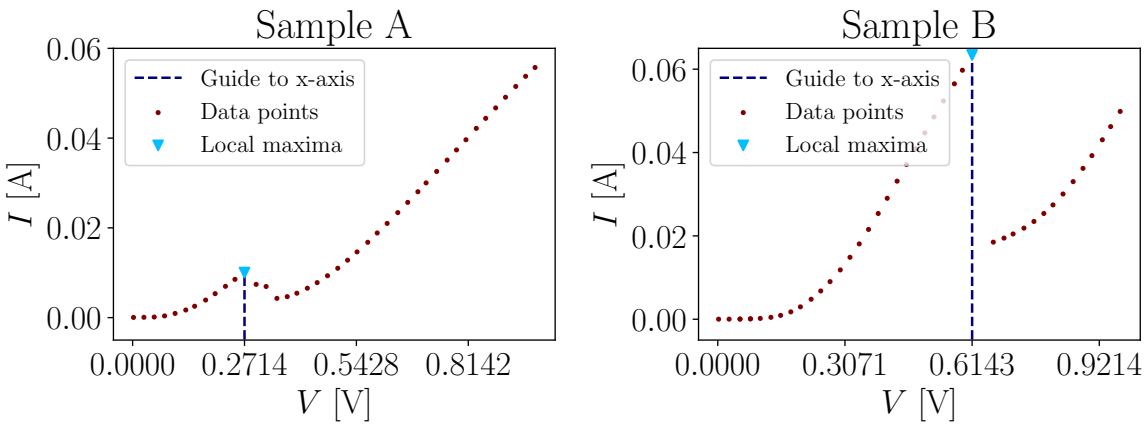


Figure 12: A plot showing the local maxima of the scatter plots of sample A and B.

4.2 Sample B

When comparing Figure 9 with Equation (5) we can tell that sample B is a double barrier. From Equation (7) and Figure 12 we can tell that $U = 614.3\,\text{mV}$. When applying Equation (8) we can conclude that the distance $b_B = 7.804\,\text{nm}$.

4.3 Sample C

By comparing Figure 10 with Equation (1) and Equation (5) there are no similarities between these equations and the plot. It can therefore be concluded that no visible quantum tunnelling occurred in sample C. With the linear regression having a slope of $1.6805 \times 10^{-2}\,\text{A}\,\text{V}^{-1}$.

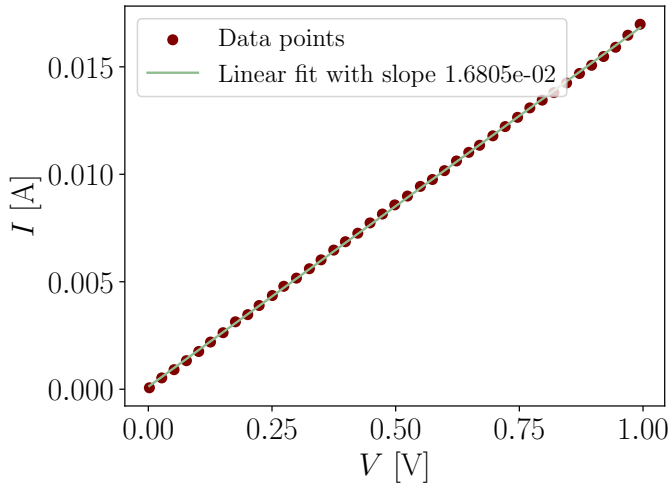


Figure 13: A linear regression applied to the scatter plot of sample C.

4.4 Sample D

When viewing Figure 11 and the scatter plot of sample D one can tel by comparison to Equation (1) that the sample is that of a single barrier. To calculate the distance a of the interposed material we can apply Equation (4) to our obtained data, that is take the natural logarithm of the current data and calculate k with the voltage data. Then by plotting a linear regression, a is obtained as the slope of the line. From Figure 14 we can conclude that the width of the innermost material in sample D is $a = 1.948\,\text{nm}$.

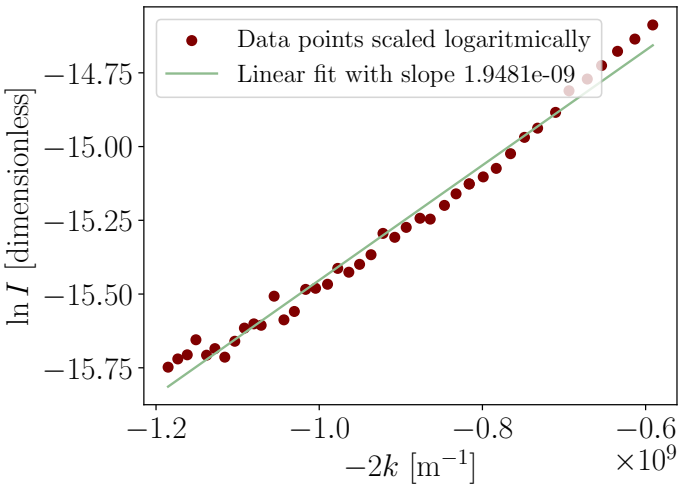


Figure 14: A linear regression applied to the scatter plot of sample D.

5 Discussion

We assumed that the incoming electrons had an energy of $E \approx 0\,\text{eV}$. The method for minimizing this energy was to submerge the samples in liquid nitrogen. Since the energy of the electron depends on the temperature in this case is $77.15\,\text{K}$ [2]. Even though this temperature is low, it is not low enough for us to say that the energy of the electrons is zero. The reason is that since the temperature is defined by kinetic energy, as long as we have a temperature above $0\,\text{K}$ the electron will have some kinetic energy. Thus our approximation in the theory that $E \approx 0\,\text{eV}$ is not correct, but will be better as $T \to 0\,\text{K}$. This has a minor effect on the result.

5.1 Sample A and B

From Figure 12 it is clear that a local maximum exists for both samples. Thus it is safe to say that the samples are both examples of a double barrier setup. It is also reasonable that increasing the distance b between the barriers would decrease the amount of energy needed for resonance. If for example, the distance between the barriers is infinite, a particle in the potential well would essentially have an infinite distance to any change in potential and may therefore be considered a free particle. A similar argument may be used for small distances as well. The wavefunction in the potential well may be described as different standing waves corresponding to different energy levels. That is, $\psi_n(x) = \sin\left(\frac{n\pi}{b}x\right)$, where n is which state the wave functions correspond to, $n = 1$ being the ground state. Then $k_n = \frac{\pi}{b}$, where k_n is the wavenumber and thus $\lambda_n = \frac{2b}{n}$. Using de Broglie's formula for momentum and wavelength an energy relation $E_n = \frac{h^2}{2m\lambda_n^2} = \frac{n^2h^2}{8mb^2}$ may be obtained, notice the similarities to Equation (8). It now becomes obvious that the energy levels increase for decreasing b. The result and analysis are therefore in line with the theory. It's also important to clarify that E_n is the energy states between the barriers and not the energy of incoming electrons.

Referring to Figure 6 it is also possible to explain why $E_1 = \frac{1}{2}\gamma eU$. Since the incoming electrons are approximated to have $E \approx 0$, and the energy floor, that is the zero-level, drops by $\frac{1}{2}\gamma eU$ it becomes clear that the electron now has E_1 energy.

In Figure 12 it is apparent that there is some kind of plateau for sample A between the local maxima and minima which is not explained by theory which predicts a sharp drop. This is not seen for sample B, but more measurements might be needed to see a similar plateau.

5.2 Sample C

Since neither Equation (1) nor Equation (5) are possible function fits for the data. When viewing Figure 13 it becomes clear that the linearity of the linear regression is due to Ohm's law $U = RI$, where U is the potential, R is the resistance and I is the current. From this formula, one can tell that the linear regression calculates the resistance of the circuit. By the Figure 13 we can conclude that $R = 16.81\,\Omega$. We can conclusively state that the resistance in the system is due to the barrier being broken, causing it to function as a resistor.

5.3 Sample D

Equation (1), and furthermore Equation (4), is only an approximation since we approximate the system to look like it did in Figure 1. Since we apply a potential bias to the system this is not the case. In Figure 5 one can see how a potential U would affect the single barrier. Thus, there means that it is an error in our approximation.

References

[1] M. Nitsch and M. B. Varga, "Quantum Tunneling Lab presentation," *Lund University, LTH, Department of Physics.*, 2023.

[2] C. Lagerström, "Laboratory Safety Manual - Liquid Nitrogen," *Örebro Univerisety*, 2022.